



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2002
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 2

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 90 minutes
- Write using black or blue pen.
- Board approved calculators may be used.
- *Each section is to be returned in a separate booklet, clearly marked Section A (Questions 1 and 2), Section B (Questions 3 and 4), Section C (Questions 5 – 7)*
Each booklet must also show your name.
- All necessary working should be shown in every question.

Total Marks - 80 marks

- Attempt **ALL** sections
- All questions are **NOT** of equal value.

Examiner: *E. Choy*

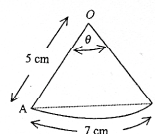
SECTION A

Question 1: [12 Marks]

- | | Marks |
|--|-------|
| (a) Use your calculator to find $e^{3.7}$ correct to 3 decimal places | 1 |
| (b) Express 300° in radians, giving your answer in terms of π | 1 |
| (c) Differentiate | |
| (i) $\tan(3x+1)$ | 2 |
| (ii) e^{4x-1} | 2 |
| (d) Find | |
| (i) $\int \sin 2x \, dx$ | 2 |
| (ii) $\int e^{-x} \, dx$ | 2 |
| (e) Find $\lim_{x \rightarrow 0} \frac{\sin 5x}{7x}$ | 2 |
- Show all working.

Question 2: [10 Marks]

Marks



An arc AB of length 7 cm of a circle of radius 5 cm subtends an angle θ at the centre O . If AB is the chord, find:

- | | |
|--|---|
| (i) $\angle AOB$ in radians. | 2 |
| (ii) The area of sector AOB . | 3 |
| (iii) The area of the segment enclosed between arc AB and chord AB . | 2 |
| (iv) The ratio of arc AB to the length of chord AB . | 3 |

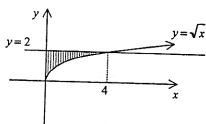
END OF SECTION A

SECTION B (Start a new booklet)

Question 3: [11 Marks]

Marks

- | | |
|--|---|
| (a) Given that $\cos 2x = \cos^2 x - \sin^2 x$ | |
| (i) Show that $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ | 3 |
| (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$ | 3 |
| (b) The diagram shows the area bounded by the y -axis, the curve $y = \sqrt{x}$ and the line $y = 2$. | 5 |

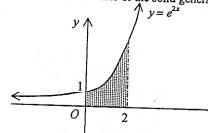


Find the area of the shaded area.

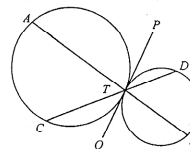
Question 4: [15 Marks]

Marks

- | | |
|--|---|
| (a) Let $f(x) = 3\cos\left(2x + \frac{\pi}{2}\right)$ | |
| (i) State the period of $f(x)$. | 2 |
| (ii) What is the range of $f(x)$? | 2 |
| (iii) Sketch the curve of $y = f(x)$, for $-\pi \leq x \leq \pi$ | 3 |
| (b) The diagram shows the region bounded by the curve $y = e^{2x}$, the x -axis, the y -axis and the line $x = 2$. The shaded area is rotated about the x -axis between $x = \frac{\pi}{2}$ and $x = 2$. Find the exact value of the volume of the solid generated. | 4 |



- | | |
|--|--|
| (c) In the diagram below, PQ is the common tangent to the two circles at T . | |
|--|--|



Copy the diagram in to your answer booklet.

Prove that AC is parallel to DB .

4

END OF SECTION B

SECTION C (Start a new booklet)

Question 5: [8 Marks]

Marks

- (i) Copy and complete the table of values for $y = \frac{1}{1+x^2}$. Give answers in exact form. 2

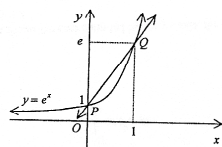
x	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2
y					

- (ii) Hence use Simpson's Rule with five function values to estimate 6

$$\int_0^2 \frac{dx}{1+x^2}$$

Question 6: [8 Marks]

Marks



The sketch above shows the curve $y = e^x$ and the points $P(0,1)$ and $Q(1,e)$ on the curve.

- (i) Show that the equation of the chord PQ is $(e-1)x - y + 1 = 0$ 4
 (ii) Find the area enclosed between the curve $y = e^x$ and the chord PQ . 4

Question 7: [16 Marks]

Marks

- (a) (i) On the SAME diagram, sketch the graphs of $y = e^{-1/x}$ and $y = 5 - x^2$, showing all intercepts with the coordinate axes. 3
 (ii) On your diagram, indicate the negative root, α , of the equation 2

$$x^3 + e^{-2/x} = 5$$

 (iii) Show that $-2 < \alpha < -1$ 2
 (iv) Use one iteration of Newton's Method, starting with $x = -2$ to show that α is approximately 3

$$-\frac{18}{e+8}$$

- (b) Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 3

- (c) Prove $7^n + 13^n + 19^n$ is a multiple of 13, if n is odd. 3

THIS IS THE END OF THE EXAMINATION

extension 1 2002: Assessment Task 2:

① a) $e^{2.7} \approx 14.880$ 3 d.p. ①

b) $180^\circ = \pi^c$

$1^\circ = \frac{\pi}{180}$

$300^\circ = \frac{300\pi}{180} = \frac{5\pi}{3}$ ①

c) (i) $3 \sec^2(3x+1)$ ②

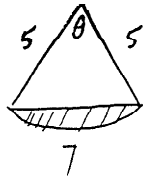
(ii) $4e^{4x-1}$ ②

d) (i) $-\frac{1}{2} \int -2 \sin 2x \, dx = -\frac{1}{2} \cos 2x + C$ ②

(ii) $-\int -e^{-x} \, dx = -e^{-x} + C$ ②

e) $\lim_{x \rightarrow 0} \frac{\sin 5x}{7x}$

$\frac{5}{7} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} = \frac{5}{7} \times 1 = \frac{5}{7}$ ②



$$(i) L = r\theta$$

$$7 = 5\theta$$

$$\theta = \frac{7}{5} = 1.4 \quad (2)$$

$$(ii) \text{ area } \overset{\text{sector}}{AOB} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 5^2 \times 1.4 = 17.5 \text{ cm}^2 \quad (3)$$

$$(iii) \text{ shaded area} = \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 25 (1.4 - \sin 1.4)$$

$$= 5.18 \text{ cm}^2 \text{ (2DP)} \quad (2)$$

$$(iv) \text{ Length ARC } AB = 7 \text{ cm}$$

$$\text{length chord } AB (x) \text{ is } x^2 = 5^2 + 5^2 - 2 \times 5 \times 5 \times \cos 1.4$$

$$x \doteq 6.44 \text{ (2DP)}$$

$$\text{ratio } \frac{\text{arc}}{\text{chord}} = \frac{7}{6.44} \doteq 1.09 \text{ (2DP)}$$

$$\text{or } \frac{350}{322}, \frac{175}{161} \text{ etc.} \quad (3)$$

SECTION B

Question 3

$$\begin{aligned}
 \text{(a) (i)} \quad \cos 2x &= \cos^2 x - \sin^2 x \\
 &= \cos^2 x - (1 - \cos^2 x) \\
 \cos 2x &= 2\cos^2 x - 1 \quad (1) \\
 \Rightarrow \cos^2 x &= \frac{1}{2}(\cos 2x + 1) \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad &\int_0^{\frac{\pi}{2}} (\cos 2x + 1) dx \\
 &= \frac{1}{2} \left[\frac{1}{2} \sin 2x + x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin 0 + 0 \right) \right] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

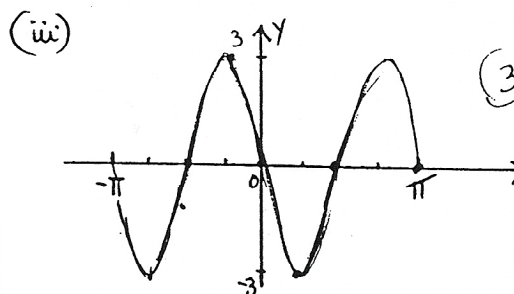
$$\begin{aligned}
 \text{Area} &= \int_0^2 y^2 dy \\
 &= \left[\frac{y^3}{3} \right]_0^2 \\
 &= \frac{8}{3} u^2
 \end{aligned}$$

$$\begin{aligned}
 \text{OR} \\
 \text{Area} &= 8 - \int_0^4 x^{\frac{1}{2}} dx \\
 &= 8 - \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4 \\
 &= 8 - \left[\frac{2}{3} (4)^{\frac{3}{2}} \right] = 8 - 5\frac{1}{3} = 2\frac{2}{3} u^2
 \end{aligned}$$

Question 4

$$\text{(a) (i) Period} = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \quad (2)$$

$$\text{(ii)} \quad -3 \leq f(x) \leq 3 \quad (2)$$



$$\text{(b)} \quad V = \pi \int_0^2 e^{4x} dx \quad (2)$$

$$\begin{aligned}
 &= \pi \cdot \frac{1}{4} \left[e^{4x} \right]_0^2 \\
 &= \frac{\pi}{4} [e^8 - e^0] \\
 &= \frac{\pi}{4} [e^8 - 1]
 \end{aligned}$$

(c) Let $\hat{ATP} = x$
 then $\hat{ACT} = x$ (angle between tan & chord = angle in alt. segment)

also $\hat{QTB} = x$ (v.o. angles)

and $\hat{TDB} = x$ (alt. segment th)

$$\Rightarrow \angle ACT = \angle BDT = x$$

$\Rightarrow$ These are equal alt. angles formed by parallel lines AC, BD

Question 5

(i) $y = \frac{1}{1+x^2}$

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
y	1	$\frac{4}{5}$	$\frac{1}{2}$	$\frac{4}{13}$	$\frac{1}{5}$

(ii) Method 1: $\int_0^2 \frac{dx}{1+x^2} = \frac{h}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2(y_2)]$
 $= \frac{1}{2} \times \frac{1}{3} \left[1 + \frac{1}{5} + 4 \left(\frac{4}{5} + \frac{4}{13} \right) + 2 \times \frac{1}{2} \right]$
 $= \frac{431}{390} = 1.10513$

Method 2: $\int_0^2 \frac{dx}{1+x^2} = \frac{1-0}{6} \left[1 + 4 \times \frac{4}{5} + \frac{1}{2} \right] + \frac{2-1}{6} \left[\frac{1}{2} + 4 \times \frac{4}{13} + \frac{1}{5} \right]$
 $= \frac{47}{60} + \frac{251}{780}$
 $= \frac{431}{390}$

Question 6

(i) Chord through $P(0,1)$ and $Q(1,e)$:

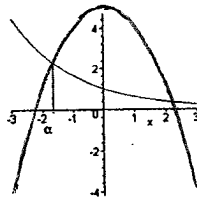
$$\frac{y-1}{e-1} = \frac{x-0}{1-0}$$

$$y-1 = x(e-1)$$

$$(e-1)x - y + 1 = 0$$

(ii) Area $= \int_0^1 [(e-1)x + 1 - e^x] dx$
 $= \left[(e-1) \frac{x^2}{2} + x - e^x \right]_0^1$
 $= \left(\frac{e-1}{2} + 1 - e \right) - (-1)$
 $= \frac{1}{2}(3-e)$

Question 7
(a) (i)



(ii) see diagram

(iii) Let $f(x) = e^{-1/2} - 5 + x^2$.

Test $f(-2) = 1.718$, $f(-1) = -0.838$

Since the function is continuous in the interval, and there is a sign change, we conclude that

α lies between -2 and -1.

(iv) Newton's Method states that if z_1 is an approximation to a root, then

$z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$ is a better approximation (under certain conditions).

Given $f(x) = e^{-1/2} - 5 + x^2$, then $f'(x) = -\frac{1}{2}e^{-1/2} + 2x$

Here $z_1 = -2$ so

$$\begin{aligned} z_2 &= -2 - \frac{e-1}{-4-\frac{1}{2}e} \\ &= -2 - \frac{2e-2}{-8-e} \\ &= -2 + \frac{2e-2}{8+e} \\ &= -\frac{18}{e+8} \approx -1.679 \end{aligned}$$

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} = 2 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2 = 2 \times 1 = 2$

(c) $P(n): 13 \mid 7^n + 13^n + 19^n$ for odd n

Let $n = 2m - 1$

$P(1): 7 + 13 + 19 = 39 = 3 \times 13$

$\therefore P(1)$ is true

Assume $P(k)$ is true ie $13 \mid 7^k + 13^k + 19^k$ for odd k

ie $7^k + 13^k + 19^k = 13R$ for some integer R

RTP $P(k+2)$ is true ie RTP $13 \mid 7^{k+2} + 13^{k+2} + 19^{k+2}$

Consider $7^{k+2} + 13^{k+2} + 19^{k+2}$

$$7^{k+2} + 13^{k+2} + 19^{k+2}$$

$$= 7^2 \times 7^k + 13^2 \times 13^k + 19^2 \times 19^k$$

$$= 7^2(7^k + 13^k + 19^k) + (13^2 - 7^2) \times 13^k + (19^2 - 7^2) \times 19^k$$

$$= 7^2(7^k + 13^k + 19^k) + 120 \times 13^k + 312 \times 19^k$$

$$= 7^2(13R) + 13(120 \times 13^{k-1} + 24 \times 19^k) \quad [\text{NB } k \geq 1]$$

$$= 13(49R + 120 \times 13^{k-1} + 24 \times 19^k)$$

$$= 13Q \quad [Q \text{ an integer}]$$

$\therefore P(k+2)$ is true if $P(k)$ is true

So by the principle of mathematical induction $P(k)$ is true for odd k .